

## Chapter 24

# The research workflow

Nine stages from a research question to knowledge that survives replication.

Statistics is a thread running through a much longer process. The diagram below names the stages of that process and links each to the parts of the curriculum that cover it in detail. No stage is optional; errors made at one are paid for at the next.

```
flowchart TD
    Q[1. Question] --> M[2. Measurements]
    M --> D[3. Design]
    D --> A[4. Acquisition]
    A --> Desc[5. Description]
    Desc --> An[6. Analysis]
    An --> I[7. Interpretation / Prediction]
    I --> V[8. Validation]
    V --> K[9. Knowledge / Decisions]
```

### 24.1 1. Question

A well-formed research question has a population, an exposure or intervention, a comparator, an outcome, and a time frame. The PICO framework from evidence-based medicine is a useful checklist. A vague question leads to a vague design and a vague answer; a well-formed question survives peer review before a single datum is collected. *Covered in:* Scientific process and research workflow, Systematic reviews and prisma.

## 24.2 2. Measurements

Every question depends on the measurement scale you choose to answer it. Accuracy, precision, reliability, and the difference between a measurement and a proxy are the vocabulary. *Covered in:* Data types tidy data accuracy and precision, Kappa icc bland altman.

## 24.3 3. Design

Design is where you choose what the data can and cannot tell you. A well-designed observational study answers some questions better than a badly designed RCT; a good RCT answers them all. *Covered in:* Observational designs and strobe.

## 24.4 4. Acquisition

Getting the data into a computer faithfully is itself a statistical problem: missingness, measurement error, and batch effects start here. *Covered in:* Import joins and missingness with dplyr, Mcar mar mnar.

## 24.5 5. Description

Before any inference, look at the data. A well-designed plot is often the whole answer. *Covered in:* Ggplot2 grammar and multi panel layouts, Descriptive statistics and table 1.

## 24.6 6. Analysis

The formal model that takes the data and returns an estimate, an interval, and a decision. *Covered in:* Courses 1–4, throughout.

## 24.7 7. Interpretation / prediction

What does the estimate mean in the scale of the science? Effect size, clinical significance, and the distinction between an average and an individual prediction live here. *Covered in:* Calibration discrimination roc auc brier score, Time dependent brier ipa external validation.

## 24.8 8. Validation

Does the finding hold in data not used to find it? External validation, nested CV, and replication pool here. *Covered in:* Cross validation nested cv bootstrap 632, Time dependent brier ipa external validation.

## **24.9 9. Knowledge / decisions**

The act of writing it down so that others can trust, cite, and build on the finding.

*Covered in:* Writing a report, Explanation vs prediction reporting.

APPENDIX · GLOSSARY

